P1. $g(E) = \sum_{\varepsilon_i} \delta(E - \varepsilon_i)$

(a) $\varepsilon_p = \frac{\vec{p}^2}{2m} \Rightarrow g_d(E) = \underset{\substack{\uparrow \\ \text{spin degeneracy}}}{2} \int \frac{d\vec{p}}{(2\pi\hbar)^d} \delta\left(E - \frac{\vec{p}^2}{2m}\right)$

$$g_{d=1}(E) = 2\int \frac{dp}{2\pi\hbar} \delta\left(E - \frac{p^2}{2m}\right)$$

Use identity $\delta(f(x)) = \sum_{x_n} \frac{1}{|f'(x_n)|} \delta(x - x_n)$

where $f(x_n) = 0$, i.e. x_n are zeros of $f(x)$.

In our case $x = p$ and $f(p) = E - \frac{p^2}{2m} \Rightarrow \frac{p_{1,2}^2}{2m} = E$, $p_{1,2} = \pm\sqrt{2mE}$

$$g_1(E) = \frac{2}{2\pi\hbar} \int_{-\infty}^{\infty} dp \left(\frac{\delta(p - p_1)}{\left|\frac{p_1}{m}\right|} + \frac{\delta(p - p_2)}{\left|\frac{p_2}{m}\right|} \right) =$$

$$= \frac{1}{\pi\hbar} \frac{m}{\sqrt{2mE}} \cdot 2 = \frac{1}{\pi\hbar}\sqrt{\frac{2m}{E}}$$

$$g_{d=2}(E) = 2\int \frac{d\vec{p}}{(2\pi\hbar)^2} \delta\left(E - \frac{p^2}{2m}\right) =$$

$$= \frac{2 \cdot 2\pi}{(2\pi\hbar)^2} \int_0^{\infty} dp \cdot p \, \delta\left(E - \frac{p^2}{2m}\right) = (u = p^2) =$$

$$= \frac{2\pi}{(2\pi\hbar)^2} \int_0^{\infty} du \, \delta\left(E - \frac{u}{2m}\right) = \frac{2\pi \cdot 2m}{(2\pi\hbar)^2} = \frac{4\pi m}{(2\pi\hbar)^2}$$

$$g_{d=3}(E) = \frac{2 \cdot 4\pi}{(2\pi\hbar)^3} \int_0^{\infty} dp \, p^2 \, \delta\left(E - \frac{p^2}{2m}\right) =$$

$$= \frac{8\pi}{(2\pi\hbar)^3} \int dp \, p^2 \frac{\delta(p - \sqrt{2mE})}{\left|\frac{p}{m}\right|} = \frac{8\pi m}{(2\pi\hbar)^3}\sqrt{2mE} = \frac{8\pi\sqrt{2}\, m^{3/2}}{h^3}\sqrt{E}$$

(b) $\varepsilon_{n,p_z} = \hbar\omega_c\left(n+\frac{1}{2}\right) + \frac{p_z^2}{2m}$ $\qquad n = 0, 1, 2 \ldots$

$$g(E) = \sum_{n,p_z} \delta\left(E - \frac{p_z^2}{2m} - \hbar\omega_c\left(n+\tfrac{1}{2}\right)\right)$$

Each Landau level is $N_\phi = \frac{B\cdot \text{Area}}{\phi_0} = \frac{eB\,L_xL_y}{2\pi\hbar c}$ degenerate

$$= \frac{eB}{mc}\,\frac{m\,L_xL_y}{2\pi\hbar}$$

$$g(E) = \frac{L_z}{2\pi\hbar}\int_{-\infty}^{\infty} dp_z\,\frac{m\omega_c L_xL_y}{2\pi\hbar}\sum_{n=0}^{\infty}\delta\left(E - \frac{p_z^2}{2m} - \hbar\omega_c\left(n+\tfrac{1}{2}\right)\right) =$$

$$= \frac{2\,L_xL_yL_z}{(2\pi\hbar)^2}\,m\omega_c\int_0^{\infty} dp_z\sum_{n=0}^{\infty}\delta\left(E - \frac{p_z^2}{2m} - \hbar\omega_c\left(n+\tfrac{1}{2}\right)\right) =$$

let $\frac{p_z^2}{2m} = z$, $p_z = \sqrt{2mz}$, $dp_z = \sqrt{\frac{m}{2}}\,\frac{dz}{\sqrt{z}}$

$$g(E) = \frac{2m\omega_c V}{(2\pi\hbar)^2}\sqrt{\frac{m}{2}}\sum_{n=0}^{\infty}\int_0^{\infty}\frac{dz}{\sqrt{z}}\,\delta\left(E - \hbar\omega_c\left(n+\tfrac{1}{2}\right) - z\right)$$

$$= \frac{\sqrt{2}\,m^{3/2}\omega_c V}{(2\pi\hbar)^2}\sum_{n=0}^{n_{max}}\frac{1}{\sqrt{E - \hbar\omega_c\left(n+\frac{1}{2}\right)}}$$

and the argument of square root is always <u>positive</u>

n_{max} is found as:

$$\hbar\omega_c\left(n_{max}+\tfrac{1}{2}\right) = E$$

$$n_{max} = \frac{E}{\hbar\omega} - \frac{1}{2}$$

(c) Limit of vanishing field $B \Rightarrow \omega_c \to 0$

$$\sum_{n=0}^{n_{max}} \frac{\omega_c}{\sqrt{E - \hbar\omega_c(n+\frac{1}{2})}} = \int_{n_{min}}^{n_{max}} \frac{dn}{\Delta n}_{=1} \frac{\omega_c}{\sqrt{E - \hbar\omega_c(n+\frac{1}{2})}} =$$

$$= \frac{2\pi}{2\pi\hbar} \int_{\hbar\omega_c n_{min}}^{\hbar\omega_c n_{max}} \frac{d(\hbar\omega_c n)}{\sqrt{E - \hbar\omega_c(n+\frac{1}{2})}} =$$

$$\left(\hbar\omega_c n_{min} = -\tfrac{1}{2}\hbar\omega_c\right)$$

$$= \frac{2\pi}{2\pi\hbar} \int_{-\frac{1}{2}\hbar\omega_c}^{E-\frac{1}{2}\hbar\omega_c} \frac{dx}{\sqrt{E - \frac{1}{2}\hbar\omega_c - x}} = \frac{2\pi}{2\pi\hbar} \left(-2\sqrt{E - \tfrac{1}{2}\hbar\omega_c - x}\right)\Bigg|_{-\frac{1}{2}\hbar\omega_c}^{E-\frac{1}{2}\hbar\omega_c}$$

$$= \frac{4\pi}{2\pi\hbar}\sqrt{E}$$

$$\Rightarrow g(E) = \frac{4\pi\sqrt{2}\, m^{3/2} V}{(2\pi\hbar)^3}\sqrt{E}$$

which differs from $g_{d=3}(E)$ in part (a) only by factor (2) due to spin degeneracy which we did not include in part (b) from the very beginning.

P2. Grand canonical for fermions

$$\Omega = -T \ln \sum_N \left\{ e^{\frac{N\mu}{T}} \sum_{(E_N)} e^{-E_N/T} \right\}$$

Ideal gas $N = \sum_k n_k$, $E_N = \sum_k \varepsilon_k n_k$

Since sum is over all N, it removes restriction on occupation numbers

$$\Omega = -T \ln \sum_{n_k} \left\{ \prod_{n_p} \exp\left(\frac{n_p \mu}{T}\right) \prod_{n_e} \exp\left(-\frac{n_e \varepsilon_e}{T}\right) \right\} =$$

$$= -T \ln \prod_p \left\{ \sum_{n_k} \exp\left(-\frac{n_k(\varepsilon_k - \mu)}{T}\right) \right\}$$

$$= -T \sum_p \ln \left\{ \sum_{n_k=0}^{\infty} e^{-\frac{n_k \xi_k}{T}} \right\} \quad , \xi_k \equiv \varepsilon_k - \mu$$

$\Rightarrow$ Fermions: $\sum_{n_k=0}^{1} \left(1 + e^{-\xi_k/T}\right)$

$$\Rightarrow \Omega = -T \sum_p \ln\left(1 + e^{-\xi_p/T}\right)$$

(includes spin degeneracy as well)

$$d\Omega = -S\,dT - p\,dV - N\,d\mu$$

$$\Rightarrow N = -\frac{\partial \Omega}{\partial \mu} = +T \sum_p \frac{e^{-\xi_p/T}}{1 + e^{-\xi_p/T}} \left(\frac{1}{T}\right) =$$

$$= \sum_k \frac{1}{e^{\frac{\varepsilon_k - \mu}{T}} + 1} \quad \left(\text{times } \underbrace{(2s+1)}_{g}\right)$$

$$N = \frac{V(2s+1)}{(2\pi)^3} \int d^3\vec{k} \; \frac{1}{e^{\frac{\varepsilon_k - \mu}{T}} + 1}$$

$$\varepsilon_k = \frac{\hbar^2 k^2}{2m} \Rightarrow k = \frac{\sqrt{2m\varepsilon}}{\hbar}$$

$$N = \frac{4\pi(2s+1)V}{(2\pi)^2} \int dk\, k^2 \frac{1}{e^{\frac{\varepsilon_k-\mu}{T}}+1}$$

$$dk = \frac{\sqrt{2m}}{\hbar} \frac{d\varepsilon}{2\sqrt{\varepsilon}} = \frac{1}{\hbar}\sqrt{\frac{m}{2\varepsilon}}\, d\varepsilon$$

$$dk\, k^2 = \frac{1}{\hbar}\sqrt{\frac{m}{2\varepsilon}}\, d\varepsilon \; \frac{2m\varepsilon}{\hbar^2} = \frac{\sqrt{2}\, m^{3/2}\sqrt{\varepsilon}\, d\varepsilon}{\hbar^3}$$

$$N = \frac{g V m^{3/2}}{\sqrt{2}\pi^2\hbar^3} \int_0^\infty d\varepsilon \sqrt{\varepsilon} \frac{1}{e^{\frac{\varepsilon-\mu}{T}}+1} = \int dN_\varepsilon$$

density of states

where $dN_\varepsilon = \frac{g V m^{3/2}}{\sqrt{2}\pi^2\hbar^3}\sqrt{\varepsilon}\, d\varepsilon$

Then energy is just $E = \int \varepsilon\, dN_\varepsilon$

$$E = C_0 \int_0^\infty d\varepsilon\, \varepsilon^{3/2} \frac{1}{e^{\frac{\varepsilon-\mu}{T}}+1} \qquad C_0 \equiv \frac{g V m^{3/2}}{\sqrt{2}\pi^2\hbar^3}$$

$$N = C_0 \int_0^\infty d\varepsilon\, \varepsilon^{1/2} \frac{1}{e^{\frac{\varepsilon-\mu}{T}}+1}$$

learn to calculate these. Use $\frac{\mu}{T} \gg 1$

let : $\varepsilon - \mu = Tz$

Then $I = \int_0^\infty d\varepsilon \frac{\varphi(\varepsilon)}{e^{\frac{\varepsilon-\mu}{T}}+1} = T\int_{-\mu/T}^\infty dz \frac{\varphi(\mu+Tz)}{e^z+1}$

$$I = T\int_0^{\mu/T} dz \, \frac{\varphi(\mu - Tz)}{e^{-z}+1} + T\int_0^{\infty} dz \, \frac{\varphi(\mu+Tz)}{e^{z}+1}$$

let $z \to -z$ for interval $\left(-\frac{\mu}{T}, 0\right)$

But $\dfrac{1}{e^{-z}+1} = \dfrac{e^{z}}{e^{z}+1} = 1 - \dfrac{1}{e^{z}+1}$

$$\Rightarrow I = \underbrace{T\int_0^{\mu/T} dz \, \varphi(\mu - Tz)}_{\text{here:}} - T\int_0^{\mu/T} dz \, \frac{\varphi(\mu-Tz)}{e^{z}+1} + T\int_0^{\infty} dz \, \frac{\varphi(\mu+Tz)}{e^{z}+1}$$

$\mu - Tz = x$

$-T\,dz = dx$

$$\Rightarrow -\int_{+\mu}^{0} dx \, \varphi(x) = \int_0^{\mu} dx \, \varphi(x)$$

replace $\frac{\mu}{T}$ by ∞ as correction is $\sim e^{-\mu/T}$

$$I = \int_0^{\mu} d\varepsilon \, \varphi(\varepsilon) + T\int_0^{\infty} \frac{\varphi(\mu+Tz) - \varphi(\mu-Tz)}{e^{z}+1} dz$$

$$\varphi(\mu+Tz) - \varphi(\mu-Tz) = 2Tz\,\varphi'(\mu) + \frac{2}{3!}(Tz)^3 \varphi'''(\mu)$$

where $\varphi'(\mu) = \dfrac{d\varphi}{d\mu}$

$$\Rightarrow I \approx \int_0^{\mu} d\varepsilon \, \varphi(\varepsilon) + 2T^2 \varphi'(\mu) \underbrace{\int_0^{\infty} \frac{z\,dz}{e^{z}+1}}_{\pi^2/12} + \dots$$

Then $\varphi_N = \sqrt{\varepsilon} \Rightarrow \int_0^\mu d\varepsilon \sqrt{\varepsilon} = \frac{2}{3}\mu^{3/2}$

$$\varphi_N' = \frac{1}{2\sqrt{\varepsilon}}$$

$$N = \frac{2C_0}{3}\mu^{3/2} + 2T^2\left(\frac{\pi^2}{12}\right)\frac{C_0}{2\sqrt{\mu}} =$$

can replace $\mu = \varepsilon_F$ in 2nd term

$$= \frac{2C_0}{3}\mu^{3/2} + \frac{\pi^2}{12} C_0 \frac{T^2}{\sqrt{\varepsilon_F}} = \frac{2C_0}{3}\mu^{3/2} + \frac{C_0\pi^2}{12}\varepsilon_F^{3/2}\left(\frac{T}{\varepsilon_F}\right)^2$$

For E $\varphi_E = \varepsilon^{3/2} \Rightarrow \int_0^\mu d\varepsilon\, \varepsilon^{3/2} = \frac{2}{5}\mu^{5/2}$

$$\varphi_E'(\mu) = \frac{3}{2}\sqrt{\mu}$$

$$\Rightarrow E = \frac{2}{5} C_0 \mu^{5/2} + 2T^2 \underbrace{\left(\frac{3}{2}\sqrt{\mu}\right)}_{\frac{3}{2}\sqrt{\varepsilon_F}}\left(\frac{\pi^2}{12}\right) C_0$$

Now, $\mu^{3/2} = \frac{3}{2}\left(\frac{N}{C_0} - \frac{\pi^2 T^2}{12\sqrt{\varepsilon_F}}\right) = \frac{3}{2}\left(\frac{N}{C_0} - \frac{\pi^2}{12}\varepsilon_F^{3/2}\left(\frac{T}{\varepsilon_F}\right)^2\right)$

$$\mu = \left(\frac{3}{2}\frac{N}{C_0}\right)^{2/3}\left(1 - \frac{\pi^2 C_0}{12N}\varepsilon_F^{3/2}\left(\frac{T}{\varepsilon_F}\right)^2\right)^{2/3}$$

$$= \left(\frac{3N}{2C_0}\right)^{2/3}\left(1 - \frac{\pi^2 C_0}{18N}\varepsilon_F^{3/2}\left(\frac{T}{\varepsilon_F}\right)^2\right)$$

But $N = C_0 \int_0^{\varepsilon_F} d\varepsilon \sqrt{\varepsilon} = \frac{2C_0}{3}\varepsilon_F^{3/2}$

$$\Rightarrow \mu = \varepsilon_F\left(1 - \frac{\pi^2}{12}\left(\frac{T}{\varepsilon_F}\right)^2\right)$$ then get $E = \frac{3N\varepsilon_F}{5}\left(1 + \frac{5\pi^2}{12}\left(\frac{T}{\varepsilon_F}\right)^2\right)$

$$E = \frac{2}{5} C_0 \, \varepsilon_F^{5/2} \left(1 - \frac{\pi^2}{12}\left(\frac{T}{\varepsilon_F}\right)^2\right)^{5/2} + 3T^2\sqrt{\varepsilon_F} \, \frac{\pi^2}{12} C_0$$

$$= \frac{2}{5} C_0 \, \varepsilon_F^{5/2} \left(1 - \frac{5\pi^2}{24}\left(\frac{T}{\varepsilon_F}\right)^2\right) + \frac{\pi^2}{4} C_0 \left(\frac{T}{\varepsilon_F}\right)^2 \varepsilon_F^{5/2}$$

Use $N = \frac{2}{3} C_0 \, \varepsilon_F^{3/2}$

$$E = \frac{3}{5} N \varepsilon_F \left(1 - \frac{5\pi^2}{24}\left(\frac{T}{\varepsilon_F}\right)^2\right) + \frac{\pi^2}{4}\left(\frac{T}{\varepsilon_F}\right)^2 \varepsilon_F \cdot \frac{3}{2} N =$$

$$= \frac{3}{5} N \varepsilon_F \left\{1 - \frac{5\pi^2}{24}\left(\frac{T}{\varepsilon_F}\right)^2 + \frac{5\pi^2}{8}\left(\frac{T}{\varepsilon_F}\right)^2\right\}$$

$$-\frac{5\pi^2}{24} + \frac{5\pi^2}{8} = \frac{-5\pi^2}{24}(1-3) = \frac{5\pi^2}{12}$$

$$\Rightarrow E = \frac{3}{5} N \varepsilon_F \left\{1 + \frac{5\pi^2}{12}\left(\frac{T}{\varepsilon_F}\right)^2\right\}.$$

$$C_V = \frac{dE}{dT} = \frac{3}{5} N \varepsilon_F \, \frac{5\pi^2}{12} \, \frac{2T}{\varepsilon_F^2} = \frac{\pi^2 T}{2\varepsilon_F} N$$